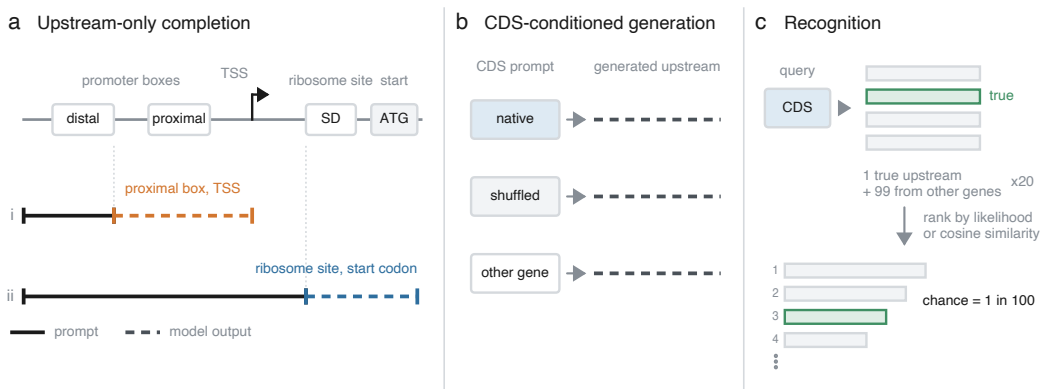
a Upstream-only completion
promoter boxes
TSS
ribosome site start
distal
proximal
SD
ATG
proximal box, TSS
i
ribosome site, start codon
ii
prompt
model output
b CDS-conditioned generation
CDS prompt
generated upstream
native
shuffled
other gene
c Recognition
query
CDS
true
1 true upstream
+ 99 from other genes
x20
rank by likelihood
or cosine similarity
1
2
3
4
chance = 1 in 100